R15

Code No: 121AF

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year Examinations, May/June - 2019
COMPUTER PROGRAMMING
(Common to CE, EEE, ME, ECE, CSE, EIE, IT, MCT, MMT, AE, AME, MIE, PTM)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a)	Give the list of basic data types in "C" language and explain.	[2]
b)	Distinguish between a keyword and a reserved word.	[3]
c)	Distinguish between a function declaration and definition.	[2]
d)	Give the list of preprocessor directives in "C" language.	[3]
e)	Why is the void pointer useful? When would you use it?	[2]
f)	Distinguish between calloc() and malloc() functions in "C" language.	[3]
g)	What are self-referential structures? Give an example.	[2]
h)	Give the syntax for bit-field declaration and explain.	[3]
i)	Distinguish between a stack and a queue.	[2]
j)	Give the applications of linked lists.	[3]

PART-B

(50 Marks)

2.a) Distinguish among 'while', 'do-while' and 'for' statements in "C" language.
b) Write a program that prints the binary equivalent of a decimal number. [5+5]

OR

3.a) Demonstrate nested if-else statements in "C" language using an example.
b) Demonstrate the usage of break and continue statements using an example. [5+5]

4. Give an overview of storage classes in "C" language. [10]

OR

5. Explain the following:
a) Multi-dimensional arrays in "C".
b) Structured programming. [5+5]

6.a) Distinguish between a constant pointer and a pointer to constant.
b) Write a "C" program to find length of a string and concatenation of two strings without using string handling functions. [3+7]

OR

7. What is a function pointer? Give the syntax and applications of a function pointer. Demonstrate how to initialize function pointers and use them in the program. [10]

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8.a) Write a program to add two complex numbers by passing structure to a function.
b) Demonstrate the following operations using examples:
i) fwrite ii) fseek [5+5]

OR

9.a) Write a "C" program to find number of lines in a file.
b) Give an overview of standard I/O in "C" language. [5+5]

10.a) Demonstrate bubble sorting using an example.
b) Write a program to reverse a word using a stack. [5+5]

OR

11.a) Write a program to demonstrate operations on a linked list.
b) Write a program to demonstrate the operations on a queue. [5+5]

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